

plot_family_trait_comparison.R

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plot_family_trait_comparison	
<i>Plot Family-Level Trait Comparison for a Focal Taxon</i>	

Description

Builds a ‘ggplot2’ log10-scale strip plot comparing the median trait value of a focal taxon against all other taxa in the same taxonomic family. Grey jittered dots represent all family members that meet the minimum record threshold; a firebrick point marks the focal taxon.

Usage

```
plot_family_trait_comparison(  
  data_family_comparison,  
  data_group_summary,  
  sel_taxon,  
  sel_domain,  
  sel_min_n = 5L,  
  graphical_options,  
  verbose = TRUE  
)
```

Arguments

data_family_comparison
A tibble of per-taxon summary statistics for all taxa in the same taxonomic family as ‘sel_taxon’, as returned by [get_family_trait_summary()]. Must contain columns ‘taxon_name’ (character), ‘n’ (integer), and ‘median’ (numeric).

data_group_summary
A single-row tibble of per-group QC statistics for the focal taxon, as produced by [generate_trait_qc_report()]. Must contain a numeric column ‘median’ used to place the firebrick reference point.

<code>sel_taxon</code>	Character scalar. Name of the focal taxon. Used in the plot title.
<code>sel_domain</code>	Character scalar. Trait domain being inspected. Used in axis labels.
<code>sel_min_n</code>	Positive integer scalar. Minimum number of records a taxon must have to appear in the plot. Defaults to '5L'.
<code>graphical_options</code>	Named list with elements 'width', 'height', 'units', 'dpi', and 'bg', as returned by 'get_active_config("graphical")'. Passed to 'ggview::canvas()'.
<code>verbose</code>	Logical. If 'TRUE' (default), the number of taxa shown in the plot after the 'sel_min_n' filter is reported via 'cli::cli_inform()'.

Details

All taxa in 'data_family_comparison' with 'n >= sel_min_n' are shown as grey jittered points on a log10 x-axis. The focal taxon's median (from 'data_group_summary') is overlaid as a larger firebrick point. When no family members meet the 'sel_min_n' threshold the grey layer is empty and only the focal taxon point is drawn.

Plot layer order follows project conventions: 'ggplot2::ggplot()' -> scales -> labels -> theme -> 'ggview::canvas()' -> geoms.

Value

A 'ggplot2' object. The plot is not printed; call 'print()' on the return value to display it.

See Also

[get_family_trait_summary()], [plot_trait_group_distribution()], [generate_trait_qc_report()]

Examples

```
## Not run:
data_family <-
  get_family_trait_summary(
    data_traits_raw = data_traits_raw,
    data_classification = data_classification,
    sel_taxon = "Anacyclus clavatus",
    sel_domain = "Leaf Area"
  )

data_summary <-
  data_qc_report |>
  dplyr::filter(
    taxon_name == "Anacyclus clavatus",
    trait_domain_name == "Leaf Area"
  )

graphical_options <-
  get_active_config("graphical")

p <-
  plot_family_trait_comparison(
    data_family_comparison = data_family,
    data_group_summary = data_summary,
    sel_taxon = "Anacyclus clavatus",
    sel_domain = "Leaf Area",
```

```
        sel_min_n = 5L,  
        graphical_options = graphical_options  
    )  
  
base::print(p)  
  
## End(Not run)
```

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